

Dynamic LION*

*LION: Empowering Multimodal Large Language Model with Dual-Level Visual Knowledge

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Abstract—In the past few years, Multimodal Large Language Models (MLLMs) have shown strong potential in expanding the modalities that the LLMs can process. The performance of LLMs with multimodal data continues to improve by leveraging the knowledge between modalities. One recent work, the dual-Level vIsual knOwledge eNhanced Multimodal Large Language Model (LION), demonstrates improvements over existing MLLMs in visual knowledge. One of the most known issues with vision models is an insufficient extraction and reasoning of visual knowledge. LION tackles this issue by incorporating visual knowledge through the progressive integration of fine-grained spatial-aware features and the use of soft prompting with high-level semantic visual evidence. Despite these advancements, the soft prompting technique remains the need for an exploration. As mentioned from the LION paper, soft prompting method still has the danger from hallucination to some degree. In this report, we propose a dynamic soft prompting to mitigate the object hallucination problem, and explore its effectiveness.

Index Terms—Multimodal Large Language Model (MLLM), LLM, Vision Encoder

I. INTRODUCTION

Recent research [1] shows an increasing use of Multimodal Large Language Models (MLLMs) with its ability to incorporate different types of knowledge across various modalities (*i.e.*, image and text). This is especially useful for vision language (VL) tasks, such as reasoning of the input image using natural language style knowledge. Nevertheless, there still exists limitations of extracting and reasoning of visual knowledge. This is largely due to most of the existing MLLMs employ vision encoder that was pretrained on a coarsely aligned image-text pairs, which ultimately leads to insufficient extraction of visual information.

One recent work, dual-Level vIsual knOwledge eNhanced Multimodal Large Language Model (LION) [2], mitigates the issue by progressively incorporates fine-grained spatial-aware visual knowledge and applies soft prompting of high-level semantic visual evidence. Specifically, the authors propose a stage-wise instruction-tuning strategy to perform image-level and region-level VL tasks. As shown in Fig. 1, it learns the visual knowledge separately by using two different adapters, and then a router module is used to aggregate the knowledge from image-level and region-level adapters into a single vision knowledge representation. In addition, LION employs a very

well-known foundation model, Recognize Anything Model (RAM) [3], as a vision extractor to obtain image tags off-the-shelf. Tags extracted from RAM are used to generate a soft prompt, which is the key component of the paper. A soft prompt is an instruction template with image tags and trainable parameters representing the prompt. The authors suggest the template as such, “According to $\langle \text{hint} \rangle$, you are allowed to use the following tags:”, where the $\langle \text{hint} \rangle$ will be replaced with a trainable soft prompt, and the tags from RAM will be inserted at the end of the template. In this way, the model can avoid potential negative influence by the imperfect predictions from RAM.

II. LIMITATIONS OF THE PREVIOUS WORK

While LION achieves high performance in image captioning, visual question answering (VQA), and visual grounding tasks, object hallucination is one of the most concerned issue from the paper, as the authors mentioned. This challenge is due to an imperfect vision model, such as RAM, generating false visual knowledge and passing it to other modalities. Because of this MLLM’s high dependency to the vision model’s performance, potential object hallucination problem is inevitable.

Not only does the performance of off-the-shelf vision models contribute to the problem, but over-reliance on statistical bias [4] and unimodal priors [5], [6] can also lead to challenges such as visual hallucinations. This shows that the challenge stems from the MLLM’s systemic nature, in which its purpose is to find correlations between modalities and align various knowledge into a unified knowledge. Nevertheless, MLLMs with foundation models are still considered to be the mainstream [7], due to their high performance and ease of implementation.

III. RELATED WORK

As discussed from Section II, extracting visual features for MLLMs is especially challenging. This is heavily due to the misalignment between text and image pairs, leading the MLLMs’ unexpected performance [8], [9]. Since it relies heavily on the dataset being used, recent works focus to utilize the text and image pairs in order to capture fine-grained visual features. NoteMR [10] utilizes knowledge notes and visual notes for reducing object hallucination.

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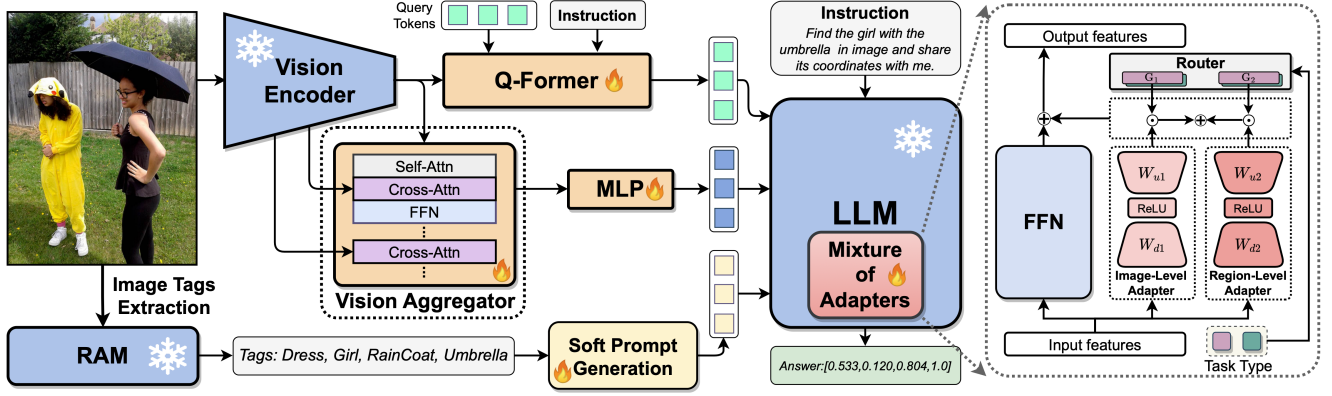


Fig. 1. Overview of the original LION [2]. The model extracts visual features from Q-Former, and combines them with fine-grained spatial-aware features from the vision aggregator. The Mixture-of-Adapters with a router in the frozen LLM dynamically fuses visual knowledge learned from different visual branches and LLM adapters based on the task types (image-level and region-level).

Y. Zeng *et al.* [11] fine-grained contrastive learning

A. Soft Prompting

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IV. METHOD

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A. Model Architecture

V. EXPERIMENTS

In this section, we present experiments including fine-tuning the BERT model for dynamic soft prompting, LION's overall training results, image-level task evaluation, and region-level task evaluation. Due to an hardware issue, we have tested the LION model with the smallest LLM model, namely Flan-T5-small. We trained our proposed model with NVIDIA RTX 4070 TI GPU with 16GB memory. It uses Parameter-Efficient-Fine-Tuning (PEFT) learning rate scale of 5.0. The training took 10 epochs, and validate every epoch.

A. Fine-tuning BERT

As discussed from Section IV-A, dynamic soft prompting requires fine-tuning process of the BERT model. The purpose of this fine-tuning is to align image captions with the given image, since most of the hallucination issues arose from the misalignment of image-text pairs. To prevent the aforementioned issue, triplet margin loss is employed which requires

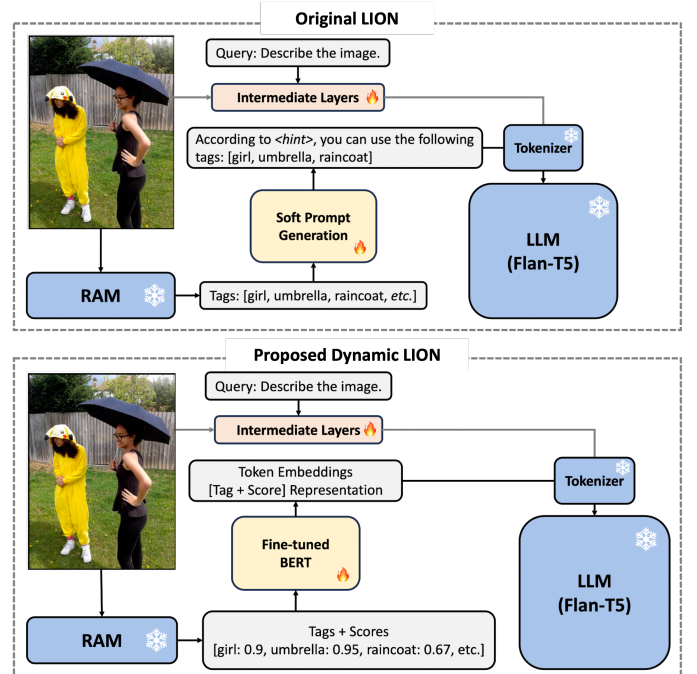


Fig. 2. Comparison between the original LION's soft prompting and the proposed dynamic soft prompting. Original soft prompting uses image tags and trainable vector (*hint*), and dynamic soft prompting uses fine-tuned BERT model to generate token embeddings, given the tags and confidence scores.

anchor, positive, and negative pairs. We use MS-COCO [12] to fine-tune the BERT model, using all five captions per image. For an anchor, RAM generates tags and confidence scores for each tag, and the token embeddings for these tags and scores will be used as an anchor from the BERT model. For positive, token embeddings of image captions are used. To be more specific, all five image captions from MS-COCO are fed to Sentence Transformer [13] to transform them into token embeddings. Then, each token embeddings are compared with the one represents the five image captions which was made by

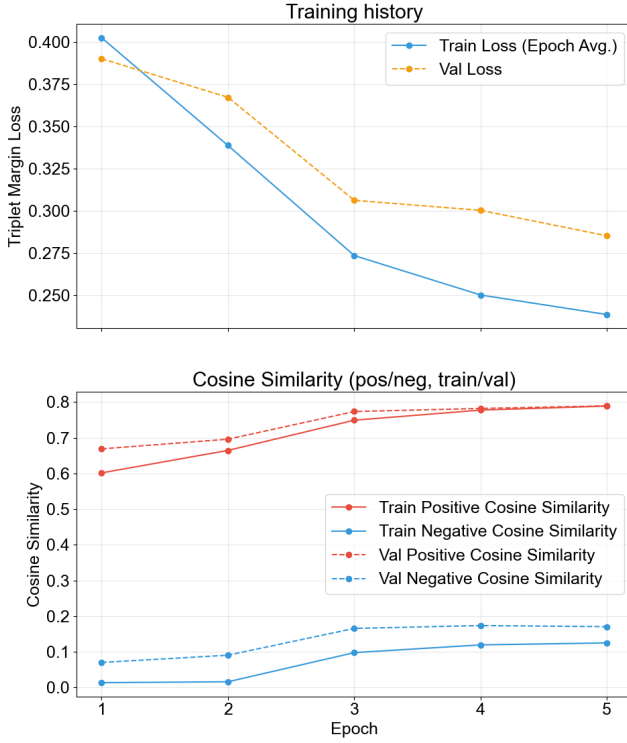


Fig. 3. Fine-tuning result of BERT model used for dynamic soft prompt generation. (Top) Both the training and validation triplet margin loss gradually decrease, showing its convergence. (Bottom) Cosine similarity scores measured from training and validation set, differentiating positives and negatives.

averaging all five token embeddings.

As shown in Fig. 3, the learning curve shows that the BERT model is successfully fine-tuned with tokenizing image tags and confidence scores with respect to MS-COCO image captions. The training loss and validation loss are decreasing gradually in a same manner, showing low loss values.

Not only does the loss is decreasing, the cosine similarity plot shows that the BERT model is actually producing an output where it is alike with positive image caption but differentiating negative image caption. This allows BERT to generate token embeddings implying image tags with image captions for dynamic soft prompt in LION.

B. Training Result of LION with Dynamic Soft Prompt

With the use of fine-tuned BERT for dynamic soft prompt, Dynamic LION is trained then evaluated with the original LION.

C. Evaluations on Image-Level VL Tasks

We evaluate the dynamic soft prompting method’s performance on two types of image-level VL tasks, which are image captioning and VQA. For image captioning, COCO caption [14] and TextCaps [15] datasets are used, and report CIDEr as the evaluation metric. Also for VQA task, OKVQA [16] and AOKVQA [17] datasets are used, and its performance is measured by Top-1 Accuracy metric.

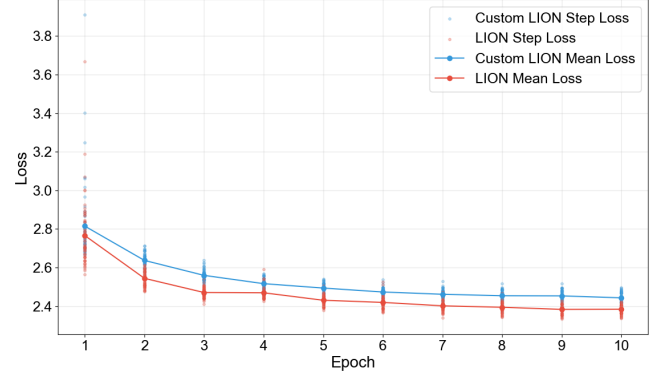


Fig. 4. Training loss curves for the original LION model (red) and the custom LION with dynamic soft prompting (blue). While the original LION achieves slightly lower loss throughout training, both models follow nearly identical learning trajectories.

TABLE I
COMPARISON OF IMAGE CAPTIONING AND VQA PERFORMANCE

Model	COCOCap	TextCaps	OKVQA	AOKVQA
LION-small	69.09	26.45	2.3%	1.8%
LION-small (ours)	21.40	4.68	16.3%	11.3%
LION-4B	138.20	104.87	51.08%	59.98%
LION-12B	139.25	108.76	57.33%	60.87%

The models used for the comparison are denoted as LION-4B, LION-12B, LION-small, and custom LION-small (ours). Each model is based off of a Flan-T5 model with different sizes of parameters. LION-4B and LION-12B are equipped with Flan-T5-XL (4B parameters) and Flan-T5-XXL (11B parameters), respectively. LION-small and custom LION-small use the same LLM, the Flan-T5-small model (80M parameters). Obviously, the higher the number of parameters the model has (*i.e.*, LION-4B and LION-12B), the more likely the model is to perform better. Our goal is to while outperform LION-small model, also achieve a certain level of performance comparable to LION-4B or even LION-12B.

As shown in Table I, LION-4B and LION-12B achieves the best performance.

D. Evaluations on Region-Level VL Task

For region-level VL task, referring expression comprehension (REC) dataset, RefCOCO [18], is used to evaluate the model’s ability to locating target object.

TABLE II
COMPARISON OF REC PERFORMANCE ON THE RefCOCO

Model	RefCOCO (IoU 0.5)	RefCOCO (IoU 0.1)
LION-small	0%	7.3%
LION-small (ours)	0 %	2.9%
LION-4B	57.57%	-
LION-12B	58.54%	-

REFERENCES

- [1] J. Kuang, Y. Shen, J. Xie, H. Luo, Z. Xu, R. Li, Y. Li, X. Cheng, X. Lin, and Y. Han, "Natural language understanding and inference with mllm in visual question answering: A survey," *ACM Computing Surveys*, vol. 57, no. 8, pp. 1–36, 2025.
- [2] G. Chen, L. Shen, R. Shao, X. Deng, and L. Nie, "Lion: Empowering multimodal large language model with dual-level visual knowledge," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 26 540–26 550.
- [3] Y. Zhang, X. Huang, J. Ma, Z. Li, Z. Luo, Y. Xie, Y. Qin, T. Luo, Y. Li, S. Liu, Y. Guo, and L. Zhang, "Recognize anything: A strong image tagging model," 2023. [Online]. Available: <https://arxiv.org/abs/2306.03514>
- [4] V. Agarwal, R. Shetty, and M. Fritz, "Towards causal vqa: Revealing and reducing spurious correlations by invariant and covariant semantic editing," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 9690–9698.
- [5] V. Gupta, Z. Li, A. Kortylewski, C. Zhang, Y. Li, and A. Yuille, "Swapmix: Diagnosing and regularizing the over-reliance on visual context in visual question answering," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 5078–5088.
- [6] Y. Han, L. Nie, J. Yin, J. Wu, and Y. Yan, "Visual perturbation-aware collaborative learning for overcoming the language prior problem," *arXiv preprint arXiv:2207.11850*, 2022.
- [7] S. Yin, C. Fu, S. Zhao, K. Li, X. Sun, T. Xu, and E. Chen, "A survey on multimodal large language models," *National Science Review*, vol. 11, no. 12, p. nwae403, 2024.
- [8] J. Cho, S. Yoon, A. Kale, F. Dernoncourt, T. Bui, and M. Bansal, "Fine-grained image captioning with clip reward," *arXiv preprint arXiv:2205.13115*, 2022.
- [9] J. Jain, J. Yang, and H. Shi, "Vcoder: Versatile vision encoders for multimodal large language models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 27 992–28 002.
- [10] W. Fang, Q. Wu, J. Chen, and Y. Xue, "guided mllm reasoning: Enhancing mllm with knowledge and visual notes for visual question answering," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 19 597–19 607.
- [11] Y. Zeng, X. Zhang, and H. Li, "Multi-grained vision language pre-training: Aligning texts with visual concepts," 2022. [Online]. Available: <https://arxiv.org/abs/2111.08276>
- [12] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft coco: Common objects in context," in *Computer Vision—ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part V 13*. Springer, 2014, pp. 740–755.
- [13] Sentence-Transformers, "all-mpnet-base-v2," <https://huggingface.co/sentence-transformers/all-mpnet-base-v2>, accessed: 2025-11-29.
- [14] X. Chen, H. Fang, T.-Y. Lin, R. Vedantam, S. Gupta, P. Dollár, and C. L. Zitnick, "Microsoft coco captions: Data collection and evaluation server," *arXiv preprint arXiv:1504.00325*, 2015.
- [15] O. Sidorov, R. Hu, M. Rohrbach, and A. Singh, "Textcaps: a dataset for image captioning with reading comprehension," in *European conference on computer vision*. Springer, 2020, pp. 742–758.
- [16] K. Marino, M. Rastegari, A. Farhadi, and R. Mottaghi, "Ok-vqa: A visual question answering benchmark requiring external knowledge," in *Proceedings of the IEEE/cvf conference on computer vision and pattern recognition*, 2019, pp. 3195–3204.
- [17] D. Schwenk, A. Khandelwal, C. Clark, K. Marino, and R. Mottaghi, "A-okvqa: A benchmark for visual question answering using world knowledge," in *European conference on computer vision*. Springer, 2022, pp. 146–162.
- [18] S. Kazemzadeh, V. Ordonez, M. Matten, and T. Berg, "Referitgame: Referring to objects in photographs of natural scenes," in *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*, 2014, pp. 787–798.